

Finding the Error in Right-Triangle Trigonometry

Answer Key

High School Geometry

Quick Answers

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|--|-----------------------------------|
| 1. (a) side opposite A = 11.18 cm, — | 4. (a) angle A = 54.62 degrees, — |
| 2. (a) side adjacent to A = 9.23 in, — | 5. (a) height = 14.78 m, — |
| 3. (a) angle A = 32.58 degrees, — | 6. (a) hypotenuse = 29.73 ft, — |
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What is verified

0 of 6 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry

HSG-SRT.C.8 – *problems 1, 2, 3, 4, 5, 6*

Difficulty 2–5 of 5 across 12 tagged checks.

Common wrong answers

- If they got 16.58: used cosine where sine was needed, pairing the opposite side with the adjacent ratio*
 - If they got -2.44: left the calculator in radian mode, so the 52 was read as 52 radians*
 - If they got 28.30: used the inverse tangent although the two known sides were the opposite side and the hypotenuse*
 - If they got 0.58: reported the ratio itself as the angle and never applied the inverse cosine*
 - If they got 7.11: left the calculator in radian mode, so the 38 was read as 38 radians*
 - If they got 14.41: used tangent, which returns the other leg rather than the hypotenuse*
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1. Hypotenuse 20 cm, $m\angle A = 34^\circ$; the student computed $20 \cos 34^\circ = 16.58$ cm for the side opposite $\angle A$. (a) Correct length. (b) Name the error.

Solution (a). The wanted side is *opposite* $\angle A$ and the known side is the hypotenuse, so the pairing is sine:

$$\sin 34^\circ = \frac{\text{opp}}{20} \quad \text{opp} = 20 \sin 34^\circ = 11.1838 \dots$$

Therefore

11.18 cm

Solution (b) — instructor-judged. A complete response says opposite-with-hypotenuse is the sine ratio, and that cosine returns the *adjacent* side instead — so 16.58 cm is the length of the other leg, not the one asked for. Award full credit only when the response names which side each ratio actually returns.

2. Hypotenuse 15 in, $m\angle A = 52^\circ$; the student entered $15 \cos 52$ and got -2.44 in for the side adjacent to $\angle A$. (a) Correct length. (b) Name the error.

Solution (a). Adjacent-with-hypotenuse is cosine, and the ratio is correct — only the angle mode was wrong:

$$\text{adj} = 15 \cos 52^\circ = 9.2349 \dots$$

Therefore

9.23 in

Solution (b) — instructor-judged. A complete response says the calculator was in radian mode, so 52 was read as 52 radians rather than 52 degrees. Award full credit only when the response also names the impossibility that exposes it: a side length cannot be negative, and no leg can exceed the hypotenuse.

3. Opposite side 7 cm, hypotenuse 13 cm; the student wrote $m\angle A = \tan^{-1}(7/13) = 28.30^\circ$. (a) Correct angle. (b) Name the error.

Solution (a). The two known sides are the opposite side and the hypotenuse, so their ratio is $\sin A$ and the angle is the inverse sine:

$$\sin A = \frac{7}{13} \quad A = \sin^{-1}\left(\frac{7}{13}\right) = 32.5789 \dots^\circ$$

Therefore

32.58°

Solution (b) — instructor-judged. A complete response says the inverse tangent belongs to the opposite-with-adjacent pairing, and the second known side here is the hypotenuse, not the adjacent side. Award full credit only when the pairing is named, not just the corrected number.

4. Adjacent side 11 m, hypotenuse 19 m; the student divided and reported $m\angle A = 0.58^\circ$. (a) Correct angle. (b) Name the error.

Solution (a). The division gives the cosine of the angle, so the inverse cosine must still be applied:

$$\cos A = \frac{11}{19} = 0.5789 \dots \quad A = \cos^{-1}(0.5789 \dots) = 54.6234 \dots^\circ$$

Therefore

54.62°

Solution (b) — instructor-judged. A complete response says the quotient is $\cos A$, not A , and the inverse cosine is what undoes it. Award full credit only when the response also gives a sanity check: two sides this close in length cannot subtend an angle of a fraction of one degree.

5. A 24 m cable at 38° to the ground; the student entered $24 \sin 38$ and got 7.11 m. (a) Correct height. (b) Name the error.

Solution (a). The height is opposite the 38° angle and the cable is the hypotenuse, so sine is the right ratio in degree mode:

$$\text{height} = 24 \sin 38^\circ = 14.7758 \dots$$

Therefore

14.78 m

Solution (b) — instructor-judged. A complete response says the calculator was in radian mode, so 38 was treated as 38 radians. Award full credit only when a rough estimate is supplied: 38° is a little over a third of a right angle, so the rise should be a little under two thirds of the 24 m cable — around 15 m, not half that.

6. $m\angle A = 29^\circ$ with adjacent side 26 ft; the student computed $26 \tan 29^\circ = 14.41$ ft for the hypotenuse. (a) Correct hypotenuse. (b) Name the error.

Solution (a). Tangent relates the two legs, so it cannot produce the hypotenuse. With the adjacent side known, cosine is the ratio, and the unknown sits underneath:

$$\cos 29^\circ = \frac{26}{c} \quad c = \frac{26}{\cos 29^\circ} = 29.7272 \dots$$

Therefore

29.73 ft

Solution (b) — instructor-judged. A complete response says tangent is the ratio of the two legs and can never return the hypotenuse, and that with the adjacent side and the angle known the hypotenuse comes from cosine with the unknown in the denominator. Award full credit only when the response also names the size check: the hypotenuse is the longest side of a right triangle, so an answer shorter than the 26 ft leg is impossible.